

Advancing Time Series Forecasting: Quantum vs. Classical LSTM Approaches

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Abstract—Time series forecasting involves analyzing historical data to predict future trends and is useful for reducing uncertainty in various phenomena. Recurrent Neural Network-based Long Short-Term Memory (LSTM) has been shown to be used for time series forecasting. However, the introduction of a hybrid quantum-classical model of LSTM, i.e., Quantum LSTM (QLSTM), opens up avenues for faster and more efficient computing and, therefore, is of significant potential. These potentials underscore the necessity for a comparative analysis of both models to assess their effectiveness and efficiency. This study evaluates the performance of QLSTM and classical LSTM models in time series forecasting. Through controlled parametric numerical experiments, the study examines the performance of both models using CPUs and GPUs in cloud environments. Our findings reveal significant improvements in the QLSTM model, with enhanced training accuracy and reduced test loss compared to the classical LSTM under similar epochs and batch sizes. The results demonstrate the superior efficiency of the QLSTM model in handling large, complex data sets compared to the classical LSTM model.

Index Terms—LSTM, QLSTM, GPU, CPU, Prediction, RMSE, Google Collab

I. INTRODUCTION

Time series forecasting uses historical data to analyze and identify patterns and predict future trends [1]. This process is pivotal for decision-making in various fields, such as finance, economics, water, and energy management. By capturing and modeling the dynamic relationships within the data, time series forecasting helps to reduce future uncertainty and provides valuable insights [2]. The use of advanced algorithms and computing technologies enhances the ability to handle large and complex datasets, making the predictions more accurate and reliable [3] leading to significant social and economic benefits [4].

Long Short-Term Memory (LSTM) network has recently emerged as a powerful variant of Recurrent Neural Network (RNN) [5]–[7]. LSTM is particularly renowned for its ability to capture long-term dependencies, making it highly effective for tasks such as time series prediction, natural language processing, and more [8]. The unique architecture, featuring gates that regulate the flow of information, allows the model to maintain and update memory over extended sequences, addressing the vanishing gradient problem commonly encountered in traditional RNN.

The Quantum Long Short-Term Memory (QLSTM) network extends the capabilities of classical LSTM by incorporating the principles from quantum computing, as shown in Fig. 1. The QLSTM leverages quantum mechanics to enhance computational power and efficiency, potentially offering superior performance in capturing the intricate patterns in complex datasets [9]. For example, QLSTM has been applied to binary classification of breast cancer histopathological images, demonstrating its ability to analyze complex medical datasets [10]. This hybrid approach aims to exploit quantum parallelism to improve the training and inference processes, thereby pushing the boundaries of what traditional LSTM architecture can achieve. An aspect of deploying LSTM and QLSTM models lies in understanding the impact of different hardware configurations on their performance. Evaluating their training efficiency and predictive accuracy on CPUs and GPUs is necessary. Optimizing performance becomes paramount in cloud computing environments, where billing is often based on the duration of computational resource usage. Minimizing execution time without compromising accuracy is essential for achieving cost-efficient solutions [11]. The following are the focus of this study:

- Evaluate the performance of the LSTM and QLSTM models in time series prediction in a cloud environment.
- Design numerical experiments on the LSTM and QLSTM models with epoch size and batch number variation to shed light on the complexity of the models in handling large complex datasets.
- Evaluate the LSTM and QLSTM models in CPU and GPU platforms regarding execution time and training efficiency.

Through our findings, we aim to advance the application of LSTM and QLSTM algorithms in real-world scenarios, promoting more efficient use of computational resources and enhancing the scalability of these advanced neural networks.

This article is structured as follows: Section II reviews the literature relevant to this study. Section III introduces the methodology and system architecture. Section IV describes the evaluation metrics. Section V describes the experimental setup in detail. The results and discussions are presented in Section VI, followed by the conclusion in Section VII. Finally, Section VIII explores potential future research directions.

II. LITERATURE REVIEW

Models that capture temporal dependencies and sequential patterns are essential for time series forecasting, especially in financial applications like predicting stock prices. Recurrent Neural Network (RNN) is specifically designed to effectively handle sequential data through directed cycles that allow information to persist over time [12]. However, RNN often faces challenges during training due to the vanishing and exploding gradient problems, which hinder its ability to capture long-term dependencies [13].

To overcome these limitations, Hochreiter and Schmidhuber introduced the Long Short-Term Memory (LSTM) model [14]. LSTM builds upon RNN by adding memory cells and gating mechanisms, i.e., input, output, and forget gates that control the flow of information and enable the network to maintain long-term dependencies. This structure helps LSTM address the vanishing gradient issues, making them highly effective for sequence-based tasks like language modeling, speech recognition, and financial time series forecasting, including stock price prediction [15].

Recent advancements in quantum computing have led to the development of a Quantum Long Short-Term Memory (QLSTM) model, which merges quantum principles with LSTM architectures. QLSTM utilizes the

computational power of quantum mechanics to perform calculations that pose challenges for classical systems [16]. By incorporating quantum circuits into the LSTM framework, QLSTM aims to enhance computational efficiency and capture complex data patterns. Preliminary research indicates significant potential for QLSTM in applications such as quantum natural language processing, quantum machine learning, and quantum-enhanced time series forecasting, such as groundwater level prediction [17]–[19].

Although QLSTM is still in the early developmental stage, studies suggest they can process complex datasets more efficiently, resulting in notable improvements in prediction accuracy and computational speed [20]. These advancements are particularly promising for financial time series forecasting, as they tackle challenges related to volatility and non-linear dependencies in stock market data. The evolution from RNN to LSTM and, more recently, to QLSTM highlights the ongoing progress in sequence modeling, offering exciting opportunities to enhance the accuracy and efficiency in time series forecasting, especially in financial domains where precision is important [21].

III. METHODOLOGY

A. Dataset

The dataset used in this study is sourced from Yahoo Finance and comprises historical stock price data for General Electric (GE) [22]. This dataset provides a robust basis for training and evaluating the performance of LSTM and QLSTM models in time series prediction.

The dataset has been structured, normalized, and divided into training and testing sets to facilitate model development and validation. The training set, which contains most of the data, is used to train both models. The testing set, reserved for the final phase, evaluates the model's predictive performance. In total, the dataset contains 3522 observations, each with five key features: open, high, low, close, and adjusted close prices. The data has been normalized before analysis.

The model utilizes 70% of all observations to ensure a comprehensive training process. For testing purposes, the last 30% of observations are used. This approach allows us to assess the model's ability to forecast future stock prices based on the historical data it has learned.

The workflow for this study, illustrated in Fig. 2, outlines the key steps undertaken, from data processing and model training to performance evaluation and prediction. This structured approach ensures a systematic and thorough analysis of the LSTM and QLSTM models

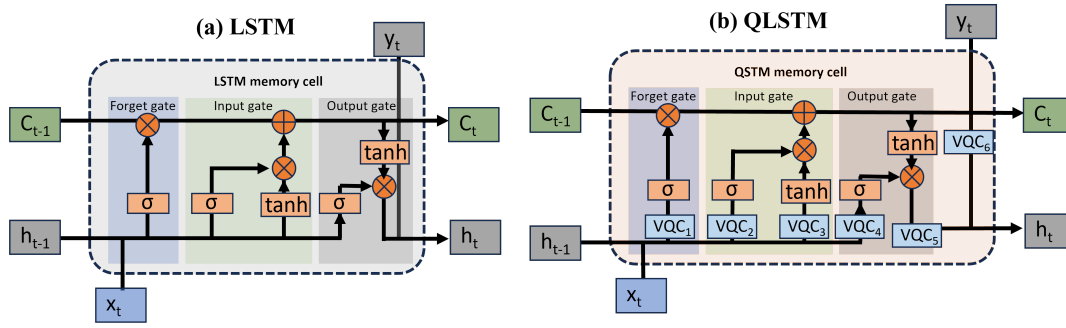


Fig. 1: (a) LSTM, and (b) QLSTM system architecture [14], [16]. The σ and $\tanh$ are the sigmoid and the hyperbolic tangent activation functions, respectively.

capabilities in time series prediction utilizing CPUs and GPUs.

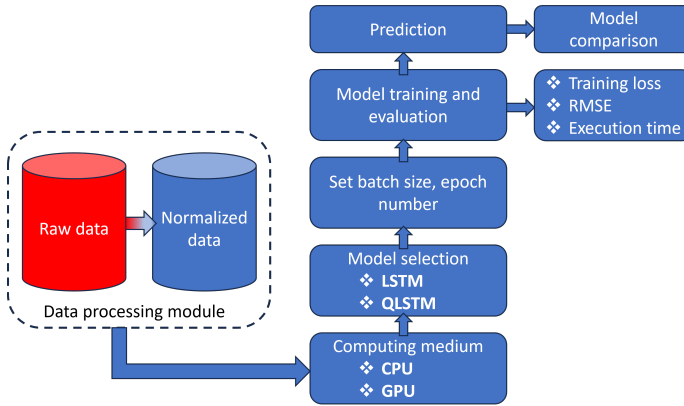


Fig. 2: Flowchart of the current study

B. System Architecture

The Long Short-Term Memory (LSTM) network, a type of Recurrent Neural Network (RNN), addresses the vanishing gradient problem through three essential gates: the Forget Gate, which manages information retention or discarding; the Input Gate, which controls new information storage; and the Output Gate, which defines the final cell output based on the current input and updated cell state, as seen in Fig. 1. This design enhances the model's ability to capture and retain key information, making LSTM valuable for time series prediction. Expanding on LSTM, the QLSTM model integrates quantum computing principles, using quantum gates and circuits to process and store the information in a quantum state. This hybrid approach uses quantum mechanics to improve computational efficiency and predictive accuracy, offering superior performance for complex sequential data tasks over LSTM.

IV. EVALUATION METRICS

To evaluate the performance of LSTM and QLSTM time series prediction models, training loss and root mean squared error (RMSE) are analyzed. Different batch sizes and epochs are used to compare CPU and GPU performance. RMSE is a benchmark statistic that compares expected and observed values, as shown in (1). It shows how effectively the model predicts the open price. The equation for RMSE is as follows:

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (1)$$

where n represents the number of predictions, y_i and $\hat{y}_i$ is the actual and predicted open price, respectively.

The computational efficiency and speed of the LSTM and QLSTM models are evaluated by analyzing the execution times on CPU and GPU platforms for different epochs and batch sizes. Batch size defines the number of samples processed per iteration, which can affect the error and training loss. An epoch is a pass through the entire training dataset. This evaluation highlights how variations in hardware and parameters, such as epochs and batch sizes, impact the speed and scalability of these models, offering insights into their optimal configuration for practical applications.

V. EXPERIMENTAL SETUP

A. Hardware

In this work, the Google collab is utilized to evaluate the performance of the LSTM and QLSTM models using benchmark CPU and GPU. The CPU configuration comprises an Intel(R) Xeon(R) processor with two cores and a clock speed of 2.20 GHz, accompanied by 12.7 GB of RAM. The GPU system features an NVIDIA L4 graphics card equipped with 7424 CUDA cores and 24 GB of memory. The current CUDA version

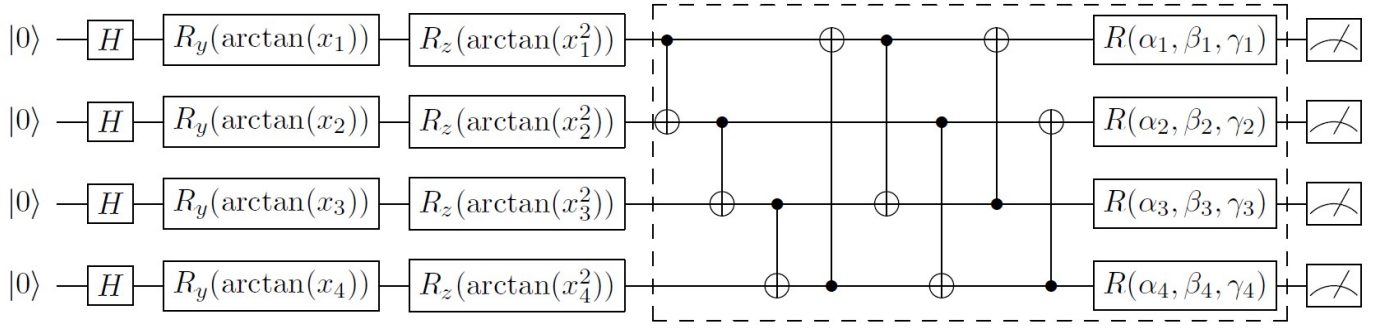


Fig. 3: Variational Quantum Circuit (VQC) framework of the QLSTM model [16], [23].

is V12.2.140, ensuring reliable hardware configuration for computational workloads utilizing CPU and GPU resources.

B. Model setup

1) *LSTM*: The LSTM model is built using a single hidden layer of 16 units, utilizing sigmoid and tanh activation functions within the LSTM cell. With a learning rate of 0.0001, the model is tuned using the Adam optimizer. Mean squared error (MSE) is used as the loss function to evaluate the accuracy of the model's predictions. The dataset utilized for training and evaluation was obtained from Yahoo Finance.

2) *QLSTM*: This study employed the identical datasets in the LSTM model to train the QLSTM model. Variational quantum layers (VQC) derived from PennyLane are utilized to substitute the LSTM layers, as illustrated in Fig. 3 [23], [24]. The configuration includes four qubits and a single variational layer and employs the Adagrad optimizer with a learning rate of 0.05. These parameters are selected to yield optimal outcomes based on experimental findings documented in the literature.

This experiment is designed to assess the performance of the LSTM and QLSTM models for time series prediction on both CPU and GPU. The architecture and parameters have been specifically tailored for this study.

VI. RESULTS AND DISCUSSION

The results and discussion comments on the numerical experiments conducted to compare the performance of LSTM and QLSTM models on both CPU and GPU platforms. Each model is assessed and evaluated based on the training loss, RMSE, execution duration, and predictive performance. The effort of this study focuses on the computational efficiency and speed achieved on both hardware platforms with accuracy. Table 1 provides a comparative analysis result for a batch size of 5 with varying epochs.

TABLE I: Comparison of LSTM and QLSTM model

CPU					GPU		
Epoch	Batch size	Training loss	RMSE	Execution time (s)	Training Loss	RMSE	Execution time (s)
LSTM							
1	1	0.335	0.37	10.39	0.335	0.37	5.18
2	1	0.0327	0.205	7.58	0.0327	0.205	5.07
3	1	0.012	0.127	6.019	0.012	0.127	5.029
4	1	0.0043	0.075	7.603	0.0043	0.075	5.12
5	1	0.0014	0.053	6.219	0.0014	0.05	5.45
QLSTM							
1	1	0.0328	0.067	1321.526	0.032	0.067	966.328
2	1	0.0013	0.046	1328.787	0.0013	0.046	971.469
3	1	0.0011	0.039	1322.435	0.0011	0.039	970.916
4	1	0.00107	0.037	1327.46	0.00107	0.037	968.14
5	1	0.00101	0.035	1315.59	0.00101	0.035	970.093

Table I demonstrates that QLSTM outperforms LSTM regarding training loss and RMSE. Although these values are consistent for both CPU and GPU, the execution time is different, with the GPU demonstrating superior computational speed relative to the CPU. The LSTM and QLSTM models show improved performance when running on GPUs compared to CPUs with similar accuracy. While QLSTM is taking a longer time to complete the computation, it can be attributed to the fact that the computation emulates a quantum computing environment developed by PennyLane [23], [24] rather than using an actual quantum computer. However, the main focus of this study is to provide a comparison in terms of model accuracy and speed for benchmarking LSTM and QLSTM models on CPUs and GPUs, which our results can do.

Fig. 4, illustrates the training loss for LSTM and QLSTM models across different batch sizes and epoch numbers. With a batch size of 1, both models showed relatively low training loss, which decreased as the number of epochs increased. However, the LSTM model experienced higher loss when the batch size increased, while the QLSTM model consistently maintained lower loss throughout all epochs. For a batch size of 10, the difference is highly noticeable in LSTM and QLSTM, demonstrating QLSTM's improved ability to handle larger and more complex datasets.

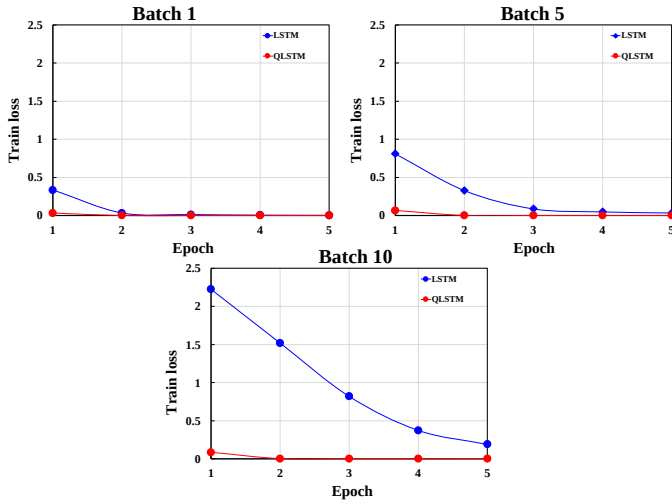


Fig. 4: Training loss comparison between LSTM and QLSTM models over different batch sizes and epochs.

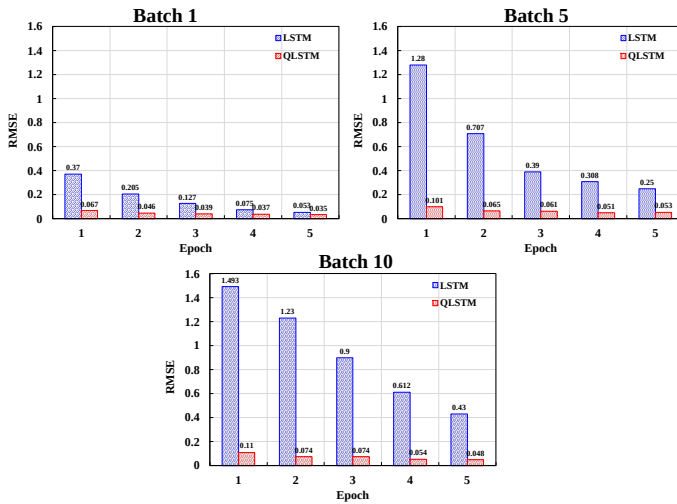


Fig. 5: Comparison of RMSE for LSTM and QLSTM models across various epochs and batch sizes.

Later, we analyzed the RMSE for both models, as shown in Fig. 5. Following a similar trend to the training loss, the QLSTM model consistently achieved lower errors for different batch sizes. In contrast, the LSTM model demonstrated higher errors, especially when processing larger batch sizes. These results further confirm the advantages of the QLSTM model, emphasizing its superior accuracy and reliability compared to the LSTM model.

To compare the performance of the LSTM and QLSTM models in CPU and GPU platforms, we compared the execution time as shown in Fig. 6 for varying batch sizes at an epoch of 5. As can be seen, both models

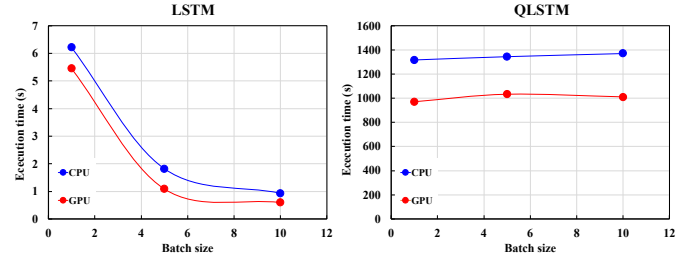


Fig. 6: Trend of the execution time of LSTM and QLSTM models on a CPU and GPU platform for different batch sizes at the epoch of five.

yield faster computation on the GPU compared to the CPU platform. With increasing batch sizes, the execution time on the CPU and GPU platforms is reduced for the LSTM model, while it remains pretty similar for the QLSTM model, though the difference increased for batch 10. This implies that GPU is a better choice for running the QLSTM model than CPU.

Following the discussion of the LSTM and QLSTM models in terms of training loss, error, and execution time, we look into the prediction capability of these models. As seen in Fig. 7, the prediction is performed using different batch sizes. It is observed that the LSTM model diverges from the original data as the batch size increases, whereas the QLSTM model stays closely aligned with it. This pattern was consistent on both CPU and GPU platforms. The comparison highlights the higher accuracy of the QLSTM model in handling complex data sets compared to the LSTM model.

The training loss, RMSE, and execution times for the LSTM and QLSTM models on CPU and GPU are evaluated. The analysis reveals improvements in prediction accuracy and efficiency using GPUs for the QLSTM model. Increasing the epoch and batch size enhances the GPU's computational speed, demonstrating its parallel processing skills. In contrast, the CPU considerably reduces training and validation durations.

VII. CONCLUSION

Our research thoroughly compared the LSTM and QLSTM models, focusing on their effectiveness in time series prediction tasks. Furthermore, we evaluated the performance of these models on both CPU and GPU platforms to assess their computational efficiency and accuracy. This assessment highlights the advantages and disadvantages of each model across various configurations. The following results can be summarized from the study:

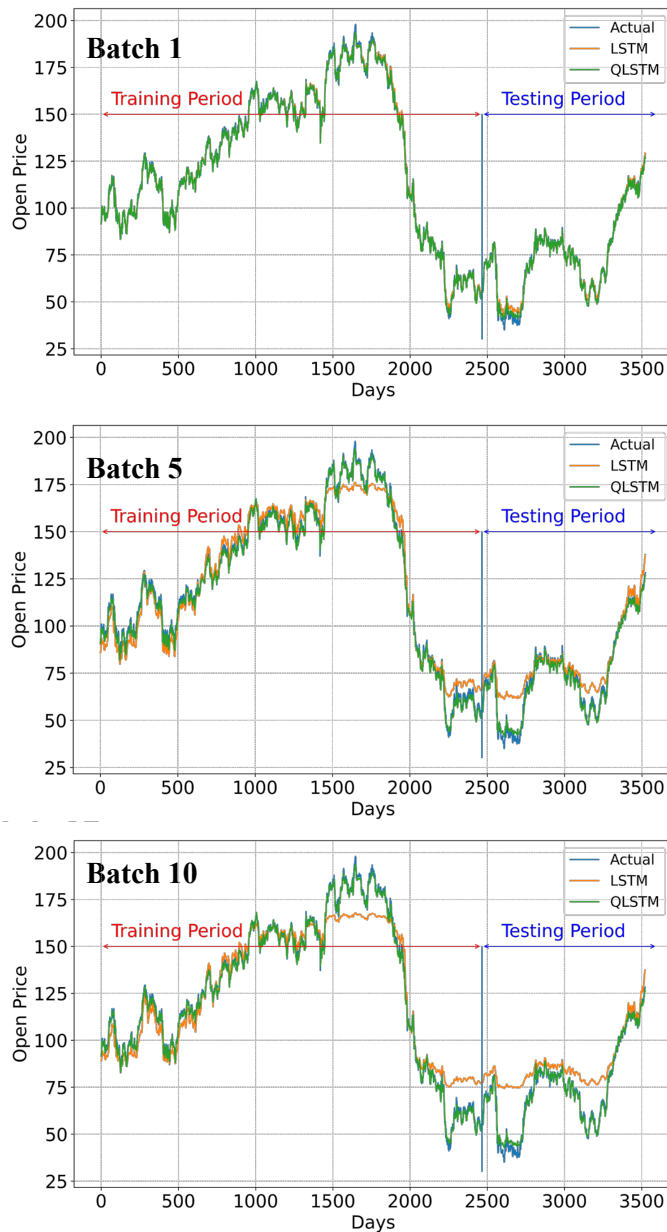


Fig. 7: Actual, training, testing, and predicted data by the LSTM and QLSTM model for different batch sizes.

- The QLSTM model showed lower train loss and RMSE across all batch sizes and epochs, indicating enhanced accuracy compared to the LSTM model.
- As batch sizes increase, predictions by the LSTM model deviate from the original data, whereas the QLSTM model consistently aligns with the original dataset.
- Both the LSTM and QLSTM models perform faster in GPU compared to CPU with increasing batch sizes and epochs.

VIII. FUTURE WORKS

Our future plan is to focus on exploring the architecture of LSTM and QLSTM models in greater detail and employing larger datasets to construct a more robust and accurate prediction model. We intend to apply these algorithms for other applications, such as groundwater level prediction, extending their applicability to real-world circumstances.

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